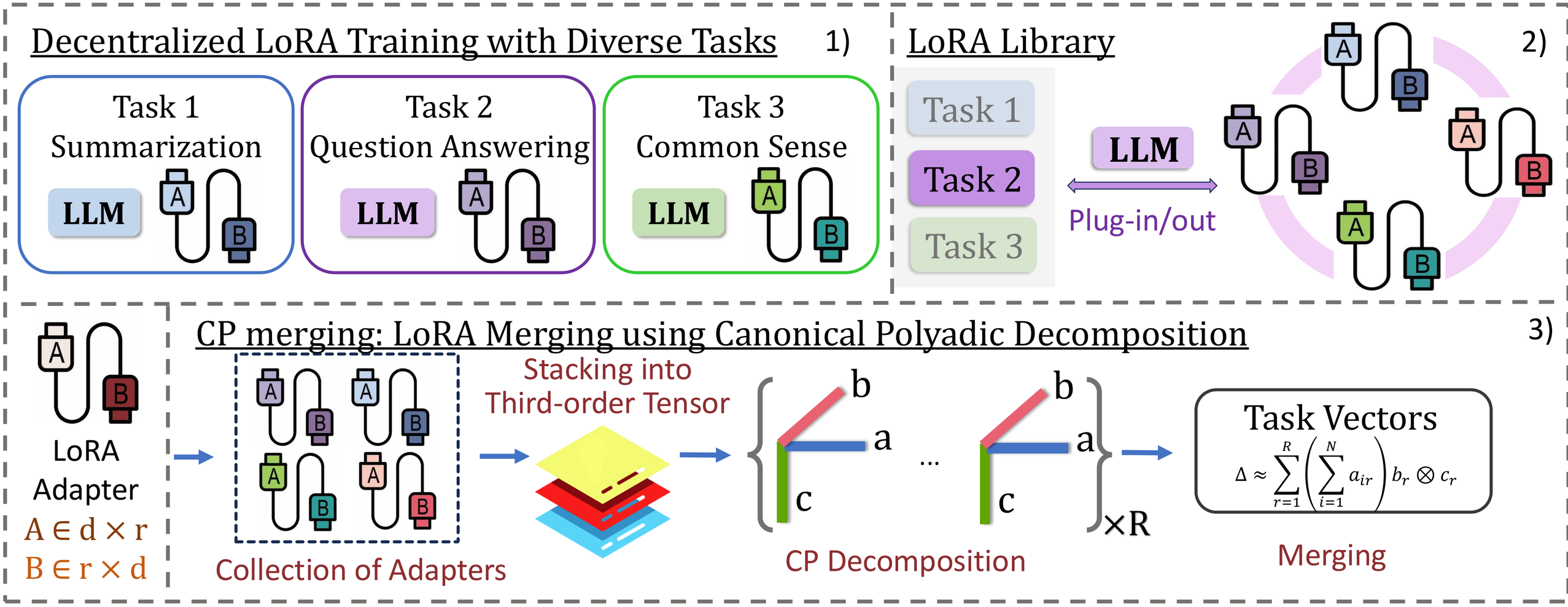

Decentralized LoRA Training with Diverse Tasks
1)
Task 1
Summarization
LLM
A
B
Task 2
Question Answering
LLM
A
B
Task 3
Common Sense
LLM
A
B
LoRA Library
2)
Task 1
Task 2
Task 3
LLM
Plug-in/out
A
B
LoRA
Adapter
A ∈ d × r
B ∈ r × d
CP merging: LoRA Merging using Canonical Polyadic Decomposition
3)
Collection of Adapters
Stacking into
Third-order Tensor
b
a
c
...
×R
CP Decomposition
Task Vectors
$\Delta \approx \sum_{r=1}^{R}\left(\sum_{i=1}^{N} a_{ir}\right) b_r \otimes c_r$
Merging